

Recording Capacity and Macroscopic Entropy in Arborescent Eternalism

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Abstract: Reversible microscopic laws leave open how the preparation required for physical recording relates to macroscopic entropy growth. This paper establishes a connection between these two features in an explicit reversible recorder. We prove that requiring the same prepared device to copy every possible message exactly constrains its initial memory to a blank state. Under an independent, uniform message distribution, we then prove that typical recording histories exhibit an extensive macroscopic entropy gain relative to that preparation at every cut in a specified late writing window. A reversible collective read interaction distinguishes memories by total occupation and physically realizes the macroclasses used in the proof. We also show how fine read access and retained writing traces change the partition, and establish that bounded reweighting preserves typical growth while unrestricted minimization can admit exceptions. The contribution is a single mechanism connecting a functional preparation requirement, typical entropy gain and physically specified read access. These local results require no global selection of the preparation or its entropy trace. They provide a recording module for Arborescent Eternalism, a framework combining one actual tenseless history with an objective structure of physically admissible alternatives. The production of the ready state, thermal relaxation and cosmological initial conditions remain further explanatory tasks.

Keywords: arborescent eternalism; reversible recording; preparation; macroentropy; typicality; temporal orientation

1 The explanatory question

How do the conditions required for physical recording connect to macroscopic entropy growth when microscopic evolution is reversible? Addressing this question requires a link between the task performed by the apparatus, its initial preparation and its subsequent dynamics. It also requires a physical account of which configurations constitute the same macrostate. Establishing these links would explain a local contribution to thermodynamic directedness.

We establish these links for a finite reversible recorder. Its task is specified before entropy is examined: one apparatus, with the same preparation, must copy any input message exactly while preserving the source. We prove that this requirement forces its memory to be blank. For an independent, uniform source, we then prove that typical prepared histories reach exponentially large occupation macroclasses at every cut in a late writing window. A collective reader measures the total number of occupied memory sites, and a reversible read interaction realizes the corresponding classes. Their sizes define the macrovolume used in the trajectory theorem. The resulting gain is measured relative to the blank preparation along individual histories.

The result connects a functional requirement to a local preparation and then to typical macroscopic growth in one device. Its physical inputs are explicit: the source distribution, the writing apparatus and its operating phase, and the reader's access to the memory. The model

describes driven recording. A thermodynamic account of relaxation, or a cosmological account of initial conditions, would require additional dynamics and preparation arguments.

The statistical-mechanical background distinguishes entropy attached to a macrostate from entropy of a fine probability distribution (Goldstein, 2001). Reversible information processing also requires separating record formation from dissipative erasure (Bennett, 1982). The present argument uses both distinctions. All entropies below are dimensionless, in nats; no heat or work bound is inferred from them.

Arborescent Eternalism (AE) places this mechanism within a description of complete histories (Mury, 2026a). An event structure organizes compatible developments; a physical projection assigns their configurations and operations; reference weights describe typicality. Actualization concerns which complete history is realized. Temporal Directedness Without an Open Future (Mury, 2026b) addresses orientation, and Global Constraints Beyond Finite Modal Data (Mury, 2026c) addresses the organization and reconstruction of alternatives. The present study supplies the local recording mechanism needed to connect preparation with entropy growth. Constraint-based accounts of complete histories offer a useful comparison (Adlam, 2022). The proofs below use the specified device and statistical law, independently of a commitment to a unique metaphysics of laws.

The central claim is that this specified module realizes a sufficient route from recording capacity to a restricted preparation and typical macroscopic growth. Its assumptions also identify where the explanation can fail. Sections 2 and 3 define the recorder and derive its required preparation. Section 4 proves the trajectory result. Section 5 realizes the read interface and tests alternative forms of access. Section 6 examines the effect of weighting and selection. Sections 7 and 8 assess the contribution to AE and its physical scope.

2 One physical module and its AE description

2.1 *Registers and reversible writing*

Let the source and memory be binary registers of equal length. A configuration is a message, a memory and an operating phase. The message is preserved during writing. The initial memory is arbitrary before the recording requirement is imposed. At stage j the device applies the controlled reversible update

$$X=(X_1,...,X_n), \quad M=(M_1,...,M_n), \quad M_j \mapsto M_j \oplus X_j. \quad (1)$$

Exclusive-or is addition modulo two. Each gate is reversible and is its own inverse: applying it twice restores the configuration. After all n gates, the memory output is

$$M_{\text{out}}=M_0 \oplus X. \quad (2)$$

An operating-phase register makes the writing schedule part of the state. At each phase, the update applies the corresponding gate and advances the phase. To reverse it, retreat the phase and apply the previous gate. Appendix A places writing in a complete reversible cycle

that also computes and uncomputes a parity output. The special preparation is therefore compatible with fully invertible microscopic evolution.

The initial operating phase is an explicit input: the reversible clock still needs initialization. The gates provide a counting model with no hidden erasure. A realistic energy conservation law, relativistic locality and an autonomous laboratory Hamiltonian remain to be supplied. Section 5 gives an effective Hermitian generator for the read interaction, which addresses observation within this model.

2.2 *Event structure and physical projection*

Within this module, E contains ordered writing events and the read events whose antecedents have been completed. A terminal read follows the last writing event. A trace protocol instead interleaves reads with writing and retains their outcomes. The physical projection Π assigns the register operations and read interactions to those events. A complete history contains the initial registers, the events and their projected configurations. All admissible initial memories belong to the model before the functional requirement is applied.

This finite module tests event access and physical projection. A controller can choose a read protocol, and Section 6 allows a goal to affect history weights. These operational roles do not establish irreducible agency or instantiate every feature of AE.

The event direction is specified independently of entropy. Reversing the whole recording cycle is possible at the microscopic level. The question here is what follows in the writing direction, conditional on the operational preparation and source law. This is the distinction between temporal directedness and thermodynamic directedness made explicit by Mury (2026b).

2.3 *Reference law and macrovolume*

The message bits are independent and balanced under a reference law w . This is a premise about the source, not a conclusion from the event tree or the recording task. For a fixed source, operating phase and any output registers held exact, the occupation observable on the memory is

$$K(M) = \sum_{i=1}^n M_i, \quad V(k) = \binom{n}{k}, \quad S_K(M) = \log V(K(M)). \quad (3)$$

The reference space contains every binary memory configuration. An occupation macroclass therefore counts all memories with the specified occupation, including those outside the capable preparation sector. Section 5 derives these classes from the collective read interface. Equation (3) states the partition whose physical realization will be established there.

The macrovolume measures how many memory configurations give the same collective reading. An observer who knows the source, preparation and reversible protocol exactly can infer the actual memory, while the collective reading still has its stated macrovolume. A physical record of the exact source-memory relation instead defines a finer partition. Measurement access, retained records and inference must therefore be distinguished.

3 Recording capacity constrains the preparation

3.1 *Exact capacity in the chosen device*

The task is universal exact copying: the same fixed preparation must produce the correct memory output for every possible message. The requirement does not mention low entropy, macrovolume, a goal score or a desired trajectory.

Proposition 1. For the device in equation (1), universal exact copying with a fixed initial memory is possible if and only if that memory is blank. The same conclusion holds for a distribution of initial memories independent of the message when copying must succeed with probability one for every message.

Proof. Equation (2) equals the input message for every message exactly when the initial memory is the all-zero string. Each initial memory in the support of a preparation distribution must meet the same condition; hence that support is the singleton blank state. Conversely, the blank state gives the required output for every message. $\square$

Under uniform counting on all memory configurations, the capable preparation occupies a fraction

$$w_M(M_0=0^n)=2^{-n}. \quad (4)$$

The functional requirement determines this small capable sector for the chosen apparatus. The implication is precise: if the device copies every message with one fixed preparation, its initial memory must lie in that sector. A process that makes or resets the apparatus is a separate physical requirement. The counting fraction describes the sector's size, not a probability that the apparatus spontaneously becomes ready.

3.2 *Why auxiliaries do not remove the resource question*

A different apparatus can use a nonblank memory, an auxiliary register or a different code. The general lesson is therefore a restriction on the complete recording resource, rather than a privileged zero string in every device.

Resource counting lemma. Suppose there are d possible messages and a recording resource with d_e possible states, consisting of a d -state displayed record and an e -state auxiliary. A fixed reversible map preserves the source and writes its value exactly in the displayed record for every message and every initial resource state in a common support of size s . Then s is at most e .

Proof. Fix a message. Reversibility maps the s distinct initial resource states to s distinct final resource states. Exact recording fixes the displayed coordinate, leaving only e auxiliary states available. Thus s cannot exceed e . The argument applies to each message, and assumes that all relevant resources have been included. $\square$

Relative to uniform resource counting, the allowed support has fraction at most $1/d$. This counting restriction is not itself a thermodynamic entropy theorem. Nonuniform reference measures, approximate recording and correlations between the source and initial apparatus

require separate assumptions and bounds. The simple recorder avoids those extensions, and Proposition 1 supplies the exact preparation used in the trajectory theorem.

The recording task gives a reason for the preparation constraint. A working recorder must have resources appropriate to its task. Explaining how physical processes supply those resources is the next question. This local functional requirement has a different scope from a cosmological boundary condition.

4 Typical growth along prepared histories

4.1 The trajectory theorem

Assume the preparation from Proposition 1 and the source law from Section 2. After t writing events, the first t memory sites contain their source bits and the remaining sites are blank. Thus the occupation on one complete microhistory is the number of ones in its first t message bits. That occupation is nondecreasing, but its macrovolume need not be: the binomial volume decreases after occupation passes the middle of the register.

For transparency, state the theorem for n divisible by four and at least eight. Define the late window, the central band and its number of cuts by

$$T=\{3n/4, \dots, n\}, \quad B=[n/4, 3n/4], \quad W=n/4+1. \quad (5)$$

Let F be the event that at least one cut in this window has occupation outside the band.

Theorem 1. In the prepared reversible recorder with independent balanced message bits,

$$w(F) \leq \min\{1, 2W \exp(-n/32)\}. \quad (6)$$

Every history outside F satisfies, at every cut in the late window,

$$S_K(M_t) - S_K(M_0) \geq \log \binom{n}{n/4}. \quad (7)$$

In particular, the failure probability tends to zero with increasing n , while the guaranteed gain has limiting density

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \binom{n}{n/4} = h(1/4), \quad h(p) = -p \log p - (1-p) \log(1-p). \quad (8)$$

Proof. The occupation at cut t is a sum of t independent fair bits. Its mean is $t/2$. For a fair bit, the centered exponential moment is a hyperbolic cosine, bounded by the exponential of one eighth of the squared argument. Independence and exponential Markov inequality therefore give

$$w\{|K_t - t/2| \geq a\} \leq 2 \exp(-2a^2/t). \quad (9)$$

The lower edge of the band is at least $n/8$ below the mean throughout the window. The upper edge is at least $n/4$ above it. A bound using $n/8$ for either tail and t at most n gives twice the exponential in equation (6) at each cut. The union bound over the W cuts proves (6). For

any occupation within the band, symmetry and unimodality of the binomial coefficients give a volume at least the coefficient in (7). The blank preparation has volume one. Finally, Stirling's formula gives (8). Appendix B supplies the exponential-moment step and an exact finite recurrence. $\square$

The theorem guarantees a gain relative to the blank preparation on each qualifying history, simultaneously at every cut in the late window. The macrovolume can fluctuate between consecutive cuts. The result concerns this writing interval; indefinite growth, thermal equilibration and the later uncomputation cycle have different requirements.

4.2 *Exact finite counts*

The simple exponential bound is deliberately conservative. At $n=64$ and $n=128$, its clipped value is one and provides no useful finite probability estimate; the smaller probabilities in Table 1 come from exact counting, not that bound. Exact counting uses integer dynamic programming over the cut, occupation and whether the band has already been missed. No random sampling or fitted entropy objective is used. The recurrence in Appendix B makes the numerical claims reproducible independently of any private repository.

Table 1 reports the probability of missing the late band at least once. A history may fall outside that band and still have a positive gain, or experience a decrease between two cuts. For every qualifying history, the displayed lower gain holds throughout the late window. The values are rounded from exact counts and binomial coefficients.

Memory size n	Fraction of blank memories	Probability of missing the late band	Guaranteed gain in nats
64	5.42×10^{-20}	0.0066538231	33.8224
128	2.94×10^{-39}	0.0003373614	69.4681
256	8.64×10^{-78}	0.0000010919337	141.1018
512	7.46×10^{-155}	0.000000000015048141	284.7138

Table 1. Conditional preparation size and exact late-window statistics for the same recorder.

The finite recurrence agrees with exhaustive message enumeration at lengths four, eight and twelve. The theorem establishes the large-size claim. The source law remains essential: an all-zero message produces no growth, although the device still has universal recording capacity. Recording capacity constrains the apparatus's preparation, while the source law controls how typical inputs fill its memory.

4.3 *What grows and what is conserved*

Writing transfers information and creates correlations. At the last writing cut, the prepared memory contains the exact message. Under a uniform source law, its reduced Shannon entropy has risen from zero to the logarithm of the number of messages. Equation (3) assigns an occupation macroentropy to each particular memory. That value depends on the message, so it must be distinguished from the entropy of the source ensemble.

The joint fine distribution over source, memory and phase has constant Shannon entropy under the reversible update. Initially it is supported on one capable configuration for each message; subsequently those equally weighted configurations remain distinct. Adding an initially pure reader and applying its reversible interaction preserves the same conclusion. Memory correlations and a large occupation macrovolume can coexist without fine entropy production.

The demonstrated growth is macroscopic growth during driven recording. Thermal relaxation would require a physical energy shell, a bath or closed-system dynamics, and suitable macroscopic variables. Here the source supplies varied information during a specified writing stage. These inputs explain the scope of the result and why reversible fine evolution can coexist with a large occupation macrovolume.

5 A physical read interaction and its access conditions

5.1 *Realizing the collective observable*

Assign the reader a pointer with $n+1$ orthogonal states. The source and memory configurations throughout the counting model are classical binary basis states. The Hilbert-space construction below embeds those states to realize a reversible read operation; it introduces no claim about a general quantum thermodynamic arrow. The read stage freezes the source and memory in their computational basis while applying a nondemolition interaction. Let the pointer dimension, Fourier matrix and pointer generator be

$$q=n+1, \quad F_{pj}=q^{-1/2}\exp(2\pi ipj/q), \quad G=F \text{diag}(2\pi j/q) F^\dagger. \quad (10)$$

Here $p,j \in \{0, \dots, q-1\}$. The occupation operator is the sum of the one-projectors on the memory sites. For interaction duration τ define

$$H_{\text{read}}=(\hbar/\tau)\hat{K} \otimes G. \quad (11)$$

The generator is Hermitian. At the prescribed duration it implements

$$U|m,p\rangle=|m,p+K(m) \bmod q\rangle. \quad (12)$$

To see this, diagonalize the pointer generator using F . The exponential phases are the Fourier eigenvalues of the cyclic pointer shift; in the memory sector with occupation k the pointer is shifted k times. A pointer prepared in a calibrated basis state consequently distinguishes every possible occupation and leaves the memory unchanged.

This effective Hamiltonian realizes reversible reading. Its uniform collective coupling and pointer-only controls define the reader's physical access. The individually addressed write ports are assumed unavailable as read ports during this phase. Equation (11) establishes that the declared interface is realizable; deriving the whole writing apparatus or a universally preferred read interface remains a separate task.

5.2 Which states this interface can distinguish

Proposition 2. For computational-basis memories, uniform collective coupling, pointer-only controls and no other accessible memory records, the maximal distinguishability classes are exactly the fibers of K .

Proof. Permuting memory sites leaves the occupation operator and read interaction invariant. Pointer-only operations preserve that symmetry. Any two configurations with the same occupation are related by a site permutation, so no protocol in this restricted class can distinguish them by the pointer. Conversely, a calibrated pointer and one read interaction distinguish all occupations. The classes therefore coincide with the occupation fibers and have the binomial sizes in equation (3). $\square$

Adaptive pointer controls preserve the same permutation symmetry as long as they remain within the declared class. Access to individual sites, source-memory relations or retained writing traces changes that class and can distinguish more states. Proposition 2 therefore justifies the macrovariable for this physical interface. An observer's complete knowledge can include additional information.

The reader has its own readiness requirement. Suppose a fixed decoder must identify every occupation exactly from a single displayed pointer result, independent of the initial pointer within its support. If that support contains s basis states, the q possible occupation shifts produce q disjoint supports of size s inside q pointer states. Hence s equals one. Any pointer basis state can serve after calibration. Relative to a uniform pointer reference, this is a deficit of $\log q$. It is a capacity requirement, not a bound on heat, work or the physical cost of resetting the pointer.

5.3 Event access and retained information

At a given past configuration, E determines which read events can occur after compatible extensions. Π specifies their actual physical record maps. The reader and its retained resources determine which results persist. For each available protocol p , write its record map as ρ_p . The realized protocol defines a partition by equality of that record. The discrimination available across the entire protocol family instead satisfies

$$z \sim z' \Leftrightarrow \rho_p(z) = \rho_p(z') \text{ for every available protocol } p. \quad (13)$$

This is the intersection of the record partitions. Adding an available protocol can only refine the family equivalence. It does not imply that all outcomes can be jointly measured or stored at no cost. In this model the read variables are classical and nondemolition; equation (13) should not be generalized to unrestricted simultaneous quantum measurements.

A controller choosing a collective record does not thereby remove an accessible fine read capability. Likewise, having a past does not mean that its complete physical information is accessible in the present. Access to a retained writing trace requires additional storage. These distinctions give E , Π and the observer's resources separate explanatory roles instead of using the phrase situated observer as a substitute for a device specification.

5.4 Relation to observational entropy

Observational entropy also connects physical measurement capabilities with macrocell volumes and probabilities (Šafránek, Aguirre and Deutsch, 2020). Its quantum formulation must additionally handle incompatible measurements (Šafránek, Deutsch and Aguirre, 2019). Equation (13) specifies these access conditions within the present module: available events, projected interactions and retained storage determine which records can distinguish histories.

For a classical partition $\mathcal{C}$ with cell probabilities p_i and reference volumes V_i , observational entropy is $S_O = -\sum_i p_i \log(p_i/V_i)$. It contains both the Shannon entropy of cell outcomes and their probability-weighted log volumes. For a determinate microstate occupying one cell, it reduces to that cell's log volume, as in equation (3). For a nontrivial source ensemble it is generally not equal to the mean of the history-wise quantity S_K alone. Our theorem and Table 1 concern individual prepared histories; they do not silently exchange these two quantities.

The construction gives an implementable record map and tests alternative interfaces within the same description of events and operations. It is compatible with observational entropy. The difference between individually addressed writing and collective reading is an explicit restriction on the apparatus during its read phase. The next section changes the read ports and retained storage to test its consequences.

5.5 Matched controls

For eight memory sites, terminal collective reading has nine classes. Exactly 254 of the 256 prepared messages have positive final occupation macroentropy: only occupations zero and eight have unit volume. The mean gain is 3.7816741154003153 nats. This endpoint statistic is distinct from the late-window criterion in Theorem 1.

Increasing the pointer capacity to 256 while keeping the collective coupling does not refine the record. At the same temporal event structure, source, preparation, writing dynamics and pointer capacity, a different projected read operation can encode the full memory configuration and produce 256 singleton classes. Order and capacity alone therefore do not specify the occupation partition.

Read protocol	Pointer capacity	Retained resource	Prepared final cell
Terminal collective	9	Final pointer	Binomial occupation volume
Terminal collective	256	Final pointer	Same occupation volume
Terminal fine	256	Final pointer	Unit volume
Writing trace	9	Additional trace memory	Unit history volume

Table 2. Paired access controls. The last row partitions histories by a retained trace, rather than measuring an instantaneous occupation macrostate.

The full trace retains the initial occupation. In a prepared history, this reading is zero and identifies the initial memory uniquely. Fixing the exact source and invertible protocol then fixes the complete history, giving a historical cell of volume one. Equation (7) concerns the instantaneous occupation partition, whereas this trace defines a partition of complete histories. Storing the trace requires an additional resource: a nine-state pointer alone cannot encode 256 alternatives injectively.

A separate resolution control gives a physical reason not to identify every small coupling perturbation with fine access. Label the eight-site binary memory by its integer code and replace the collective analog value by occupation plus that code divided by 1024. Exact measurement resolves all configurations. Rounding to the nearest integer still returns occupation for every configuration if additive read error is bounded by $1/8$: the perturbation plus error remains strictly between $-1/2$ and $+1/2$. This is a robustness result for declared resolution and noise, not for arbitrary detectors or partitions chosen after inspecting the gain.

The event executor and an independent causal register implementation agree on every record for every source and every initial memory across the four protocols in Table 2. The correspondence is exact in the finite model. A separate four-site check confirms that the 24 orders of independent writes do not create 24 physical alternatives: all produce the same final configuration for each of the 16 source messages. This quotient control concerns independent writes, not the fixed schedule used for the entropy window.

6 Typicality and global selection are different claims

6.1 *Bounded weighting preserves a typical result*

The preparation requirement and trajectory theorem use the device and reference source law. A further weighting or selection rule raises two questions: does the altered probability law preserve the typical result, and which complete history is actual? The first has a quantitative answer under a density bound. The second requires a rule of actuality.

Proposition 3. If a probability law P on prepared histories has density at most Λ relative to w , its late-window failure probability is at most Λ times the reference failure probability. More generally, if P assigns at most δ probability outside the capable sector L and its conditional density within L is at most Λ relative to the reference law conditioned on L , then

$$P(L^c \cup F) \leq \delta + \Lambda w(F|L). \quad (14)$$

Proof. Integrate the bounded density over the failure event inside L . Its conditional probability is at most the stated reference multiple. Add the probability outside L , using the factor $P(L)$ at most one for the contribution inside. $\square$

For the fixed recorder, the source law already describes prepared histories, so L need not be selected by a second score. Equation (14) is useful for extensions in which readiness can fail or weights can favor some messages. Its content is preservation under a controlled change of law; it does not derive such a law.

For a concrete goal-dependent example, let the parity output be required to match a binary goal g through an observation channel of fixed error e between zero and one half. Give a message likelihood $1 - e$ when its parity matches g and likelihood e otherwise. Both source parities have probability one half, so normalization gives

$$P_{g,e}(X) = 2w(X)l_e(X,g), \quad \frac{dP_{g,e}}{dw} \leq 2(1-e) < 2. \quad (15)$$

This weighting favors the required parity while increasing the failure-risk bound by a factor smaller than two. The factor follows from the stated channel error. The example gives a goal-sensitive statistical law, and it can equally be read as conditioning the source on a noisy observation. Establishing irreducible agency or a fundamental actuality law would require further premises.

6.2 *Minimization does not supply the same guarantee*

An evaluator formed as negative log reference weight plus negative log likelihood has a different interpretation when its global minimum is selected. With uniform source weights, every message of the preferred parity is a minimizer. For goal zero, the all-zero message is one of them and writes no occupied sites. For goal one, a message with only one occupied site is a minimizer and misses the declared central late band when n is at least eight.

The minimizing evaluator admits histories that satisfy the trajectory property and histories that fail it. A uniform rule for breaking ties would have density at most two and would preserve a typicality bound. Such a rule is an additional actualization premise; the set of minimizers alone does not supply it.

AE can use J in a proposal for whole-history actualization. Its variational condition must then be assessed separately from event orientation, functional preparation and macroscopic growth. The local writing mechanism already produces the conditional increase. A law over alternatives determines their probabilities; guaranteeing the trajectory of one actual history requires control of how that history is selected.

7 The explanatory contribution of AE

7.1 *What has been unified*

The positive result is that one specified AE module supplies a coherent explanatory chain. Event dependencies locate preparation, writing and read stages. Π makes their operations physically explicit. The recording task restricts the preparation of the same memory whose occupation is subsequently measured. The source weights yield a quantitative trajectory-typicality statement. A controlled goal weighting can modify history statistics without destroying the bound. None of these roles is performed by an entropy reward hidden inside J .

The controls identify what each part contributes. Without universal copying capacity, the blank preparation no longer follows. An all-zero source suppresses growth. Fine read access removes occupation multiplicity, and a retained trace changes the partition of histories. Moving

the read event before writing changes the observed configuration. These changes locate the explanation in the task, statistical inputs, event access and physical interactions.

The companion studies address complementary tasks. The orientation study (Mury, 2026b) gives sufficient bridge conditions and a symmetry obstruction. The global-admissibility study (Mury, 2026c) examines the organization of finite records and complete alternatives, including equivalent serializations of independent events. This study adds a mechanism connecting a functionally constrained local preparation with an oriented interval of typical macroscopic growth. A common physical realization must ultimately justify how the requirements of the three studies are supplied together.

7.2 *Why a matched causal account is compatible with this contribution*

AE is used here as an explanatory theory with causal physical realizations. Event dependencies and projected interactions specify what can influence a later record. A causal implementation preserving the task, admissible events, operations, source law, read access and retained resources realizes the same module. Section 5.5 checks this correspondence in the finite device. AE further treats admissible alternatives as objectively organized and posits one actual complete history. Its variational formulation concerns global evaluation and actualization. Those wider commitments are independent of the local proofs: a causal description can share the mechanism while interpreting alternatives instrumentally or leaving actuality open (Mury, 2026a, Section 10.1).

The explanatory test is whether the framework makes the physical conditions explicit, connects them consistently and supports counterfactual comparisons. Changing the source, read interaction or retained record changes the conclusion while other components stay fixed. Preserving the physical package preserves the result under a causal description. These controls show how the mechanism works and which assumptions it needs.

The correspondence applies to this specified structure and physical content. It leaves general causal models and AE's wider ontology distinct. The contribution established here is a finite construction that integrates preparation, dynamics and access; Section 7.3 compares it with established explanations. An additional empirical advantage remains to be demonstrated.

7.3 *Relation to existing explanations*

Special conditions for record formation have a long history. Reichenbach discusses registering instruments in *The Direction of Time* (1956). Albert examines asymmetries of knowledge and intervention in *Time and Chance* (2000), Chapter 6 (pp. 113–130). Past-Hypothesis accounts connect local records with a special cosmological boundary and statistical dynamics (Albert, 2000; Loewer, 2012). Here the implication begins with an independently specified copying task: for a fixed device to perform it, the initial memory must be restricted. This explains the local readiness requirement conditional on functioning. A physical account of how that readiness is produced remains necessary.

Mlodinow and Brun (2014) analyze reversible, nondissipative memories and require preparation that is not fine-tuned to one particular recorded state. Their argument addresses

alignment with a robust thermodynamic arrow in the surroundings. Proposition 1 imposes one fixed preparation for every message and proves that the capable state of this XOR recorder is unique. The trajectory theorem then tracks that memory's occupation macroentropy during writing. Its conclusion concerns the recorder rather than a psychological arrow or an ambient thermodynamic arrow.

Landauer (1961) and Bennett (1982) distinguish reversible recording from dissipative erasure. The resource lemma uses reversibility to count how many initial states can yield an exact displayed record. It supplies a capacity restriction; a heat-cost bound would need thermodynamic assumptions. Appendix A keeps subsequent uncomputation explicit in the reversible cycle.

The added content is a combination established for one device. The copying task constrains its preparation; a uniform source gives a finite bound along individual histories; a reversible read interaction realizes the macrocells; and bounded reweighting preserves the typical result. Each ingredient has precedents. Their combination makes the local explanation explicit and reproducible, while leaving cosmological extension and comparative empirical advantages open.

A local recorder can obtain usable resources from its environment. Repeated recording requires an account of how those resources are supplied, consumed or reset. The reversible cycle makes the local entropy trace clear: writing can produce a large macrovolume, and subsequent uncomputation can return the device to its earlier configuration.

An actual tenseless block can contain preparation, writing and uncomputation in one complete history. The finite mechanism establishes a directional result within its writing phase. Connecting that result to the much longer cosmological arrow requires a physical explanation of the preparation and statistical conditions on the larger scale.

8 Objections and conclusion

The low preparation has simply been assumed. Proposition 1 derives the blank state from the task and apparatus. The reader's exact display task similarly constrains its calibrated preparation. These are conditional requirements for functioning devices. Producing the ready states is a further physical question.

The entropy increase is an artifact of ignoring correlations. Source-memory correlations remain present under reversible writing. The specified collective reader distinguishes occupation classes; fine access to source-memory correlations defines a different partition and entropy. The controls test that change of access explicitly. The entropy assignment is tied to the stated physical interface and retained records.

Recording is being mistaken for thermal relaxation. The result describes driven writing under balanced inputs and proves an extensive occupation macroentropy gain along typical histories. A thermal account would also need temperature, energy transfer and appropriate equilibrium dynamics. These physical requirements determine how the model could be extended.

AE has not been shown necessary. The result supplies a concrete local explanation with explicit inputs and causal realizations. A matched causal description expresses the same module. The proof establishes the construction's sufficiency; necessity, empirical discrimination and a cosmological explanation remain separate research obligations.

Selection should guarantee the actual arrow. Proposition 3 bounds the failure probability for a law whose density is controlled. The parity example shows that global minimization alone can admit exceptional histories. A guarantee for the actual history requires an additional condition on its selection. Orientation, statistical typicality and actualization therefore have distinct roles within the architecture.

One reversible mechanism connects a required preparation with typical macroscopic growth. Universal copying forces the capable memory state. Under an independent balanced source, most prepared histories reach large occupation macrovolumes at every cut in a late writing window. A reversible collective read interaction supplies the physical partition, and the access controls identify when a different partition results. Further weighting preserves the probability bound when its density is controlled.

Within AE, this construction shows how event dependencies, physical projection, recording capacity and source weights work together in a local explanation. The next physical task is to justify the apparatus and access regime in a wider domain, then connect their preparation and statistical conditions to cosmology. A common realization compatible with the orientation and global-admissibility studies would test the proposed reuse of AE's structure across explanations.

Appendix A A complete reversible cycle

Add a binary parity output a and a phase c with $2n + 2$ values. A full cycle performs n writing gates, toggles a by the parity of the resulting memory, reverses the n writing gates, and toggles a by the parity of the exclusive-or of the restored memory and source. Advance the phase after each operation. Every phase operation is an involution. The inverse cycle retreats the phase and applies its previous operation.

For arbitrary initial memory and output, the first toggle adds the parity of the initial memory exclusive-or source. The second toggle adds exactly the same bit after memory restoration. They cancel. Consequently the whole cycle restores source, memory, output and phase for every initial configuration, not only capable preparations. With blank memory and output initially zero, the intermediate output faithfully displays source parity. These output and phase initializations are distinct requirements from Proposition 1.

The writing portion is the interval studied in Theorem 1. During the reverse writing portion occupation returns toward the blank state in capable histories. A cyclic model cannot support an indefinitely increasing entropy assignment. Its role is to establish that the local writing theorem is compatible with an exactly reversible completed physical history.

Appendix B Concentration and exact reproduction

B1 The exponential moment

For one fair bit b , the centered moment is $\cosh(\theta/2)$. The inequality $\log \cosh u \leq u^2/2$ follows by differentiating: $\tanh u \leq u$ for $u \geq 0$, and both sides agree at zero; symmetry handles $u < 0$. For t independent bits this gives

$$\mathbb{E}_w \exp\{\theta(K_t - t/2)\} \leq \exp(t\theta^2/8). \quad (16)$$

Applying Markov inequality and minimizing at $\theta = 4a/t$ yields the upper-tail bound $\exp(-2a^2/t)$. Negative argument gives the lower tail. Their sum gives equation (9). This is the fair-bit special case of a standard bounded-variable concentration inequality (Hoeffding, 1963), derived here so the proof is self-contained.

B2 Counting the late-window failures

Let D at cut t , occupation k and flag b count message prefixes, where b is zero until the late band has been missed and one thereafter. Initialize the count at cut zero, occupation zero and flag zero to one; all other entries are zero. For each next input bit u equal to zero or one, add the current count to the state with occupation k plus u and flag

$$b' = b \vee [t+1 \geq 3n/4 \wedge k+u \notin [n/4, 3n/4]]. \quad (17)$$

The final sum of entries with flag one, divided by the number of messages, is the exact failure probability. Splitting that sum by final occupation parity supplies exact counts for equation (15). Integer arithmetic gives the counts; logarithms of binomial coefficients give the gains in Table 1. For sizes not divisible by four, replace the window start and band endpoints by the corresponding ceiling and floor, as in the implementation below. The main theorem intentionally avoids those rounding conventions.

The following complete Python routine reproduces the reference failures and gains in Table 1 without simulation. It can be run with Python 3 using only the standard library.

```
from fractions import Fraction
from math import comb, log

def late_counts(n):
    begin = (3*n + 3)//4
    lo, hi = (n + 3)//4, 3*n//4
    counts = {(0, False): 1}
    for t in range(1, n + 1):
        nxt = {}
        for (k, failed), count in counts.items():
            for bit in (0, 1):
                j = k + bit
                f = failed or (t >= begin and not lo <= j <= hi)
                nxt[j, f] = nxt.get((j, f), 0) + count
        counts = nxt
    bad = [sum(c for (k, f), c in counts.items()
               if f and k % 2 == p) for p in (0, 1)]
```

```
return bad
```

```
for n in (64, 128, 256, 512):  
    bad = late_counts(n)  
    risk = Fraction(sum(bad), 1 << n)  
    print(n, float(Fraction(1, 1 << n)), float(risk),  
          log(comb(n, (n + 3)//4)))
```

For a rational channel error e , the goal-conditioned failure probability is the sum of $1 - e$ times the matching-parity failure count and e times the opposite-parity failure count, divided by the number of messages of one parity. This is exact rational arithmetic and reproduces the bound in Section 6.

B3 Read controls

For the eight-site endpoint, enumerate integers from zero to 255 as memory states. Group them by their number of set bits; the class sizes are the binomial coefficients. Sum the logarithms of those sizes over all source messages and divide by 256 to obtain the collective mean reported in Section 5.5. Grouping instead by the full integer gives singleton classes. To reproduce trace classes, fix a source, enumerate every initial memory, apply successive exclusive-or updates and record occupations before writing and after each gate. A prepared history's initial record zero singles out its initial memory. The event and causal descriptions implement these same explicitly stated maps.

Research transparency

The essential proofs and finite counting algorithm are included in this manuscript. The public supplement is fixed to commit 0b30398bf80aa72fd47a040bf124080c9f4a1695: https://github.com/Legiolab/arborescent_eternalism_public/tree/0b30398bf80aa72fd47a040bf124080c9f4a1695/proofs/paper-c-recording-entropy/. The `event_access_controls/check.py` script and `results.json` verify agreement between the event executor and an independently implemented causal recorder, and all 24 orders of four writes (Section 5.5). The `physical_read_controls/check.py` script and `results.json` provide the companion physical read controls. These controls do not replace the in-text proofs or extend them to a cosmological preparation.

Generative AI systems assisted conceptual discussion, formalization, code development, literature discovery and manuscript drafting. Numerical claims were produced by deterministic Python computations, not certified by language-model arithmetic. AI systems are not evidential sources or authors. Responsibility for reviewing and approving the submitted arguments, citations and manuscript remains with the author.

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